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connection
to span of
a set?

not necessarily
minimum set?
or always?

useful for finding finite
open covers for
compactness?

Can open sets be
convex? I think so

Hyperplane is plane of

$$\{x : a^T x = b, a \in \mathbb{R}^n, b \in \mathbb{R}\}$$

such that they divide
space in half

therefore
always
of dim
 $n-1$

Divides $\mathbb{R}^n$

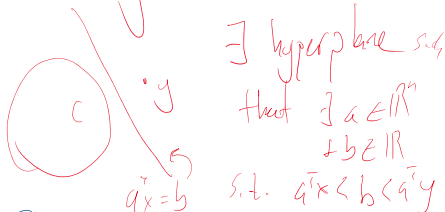
$$\{x : a^T x < b\} \text{ lower halfspace?}$$

$$\{x : a^T x > b\} \text{ upper halfspace?}$$

Separating hyperplane (strict separation)

convex set C closed

+ $y \notin C$,



$\exists$ hyperplane s.t.
that $\exists a \in \mathbb{R}^n$
+ $b \in \mathbb{R}$
s.t. $a^T x < b < a^T y$
 $\forall x \in C$

where is
vector x
in diagram?

if C not closed,

y could be on
boundary of C ,
but then no separating

hyperplane except one

that goes through y
(tangent to boundary C ?)

$$\text{co}(C) = \bigcap \{D \subseteq \mathbb{R}^n : C \subseteq D \text{ and } D \text{ is convex}\}$$

If D is any convex set satisfying $C \subseteq D$, then $\text{co}(C) \subseteq D$ by definition.

2. Open Sets and Convexity Open sets can be convex. Standard examples in $\mathbb{R}^n$ include:

- Any open ball: $B_r(x_0) = \{x \in \mathbb{R}^n : \|x - x_0\| < r\}$
- Open half-spaces: $\{x \in \mathbb{R}^n : a^T x < b\}$
- The interior of any convex polyhedron or simplex

3. Connection to the Span of a Set

The linear span $\text{span}(C)$ and the convex hull $\text{co}(C)$ represent two levels of algebraic restriction:

- $\text{span}(C)$: All linear combinations $\sum \alpha_i x_i$ with $\alpha_i \in \mathbb{R}$ (unconstrained).
- $\text{co}(C)$: All convex combinations $\sum \lambda_i x_i$ with the simplex constraints $\lambda_i \geq 0$ and $\sum \lambda_i = 1$.

Because the convex coefficients are a constrained subset of arbitrary linear coefficients, the convex hull is always a subset of the linear span:

$$\text{co}(C) \subseteq \text{aff}(C) \subseteq \text{span}(C)$$

(where $\text{aff}(C)$ is the affine hull, requiring only $\sum \lambda_i = 1$ without non-negativity).

4. Utility for Finding Finite Open Covers in Compactness The convex hull itself is generally not used to construct topological open covers for the Heine-Borel test. Convex hulls preserve boundedness, but they do not preserve openness or closedness automatically in infinite dimensions, and open covers typically consist of $\{B_\epsilon(x)\}_{x \in K}$.

However, convex hulls are central to compactness in two foundational ways:

- Preservation of Compactness:** In $\mathbb{R}^n$, if a set K is compact, its convex hull $\text{co}(K)$ is also compact.
- Fixed Point Theorems & Finite Sub-simplices:** In functional analysis and game theory (e.g., the Kakutani or Brouwer Fixed Point Theorems), compact convex finite open covers of simplices formed by convex hulls of discrete points (via Sperner's Lemma and nerves of coverings).

From <https://www.youtube.com/watch?v=3G8H8H8H8H8>

2. Separating Hyperplanes

A hyperplane takes the linear algebraic form $a^T x = b$, with $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$. A half-space is characterized by the inequality $a^T x \leq b$ or $a^T x \geq b$.

Analytical Extension on Half-spaces:

A hyperplane constitutes the precise geometric boundary of a closed half-space. The mapping between the two is direct.

Separating Hyperplane Theorem:

If C is a closed convex set and $y \notin C$, there exists a strictly separating hyperplane such that $a^T x < b < a^T y$ for all $x \in C$.

Analytical Extension on Boundary Conditions:

If the vector y resides on the boundary of C , strict separation fails. Instead, a supporting hyperplane tangent to the boundary of C at y is established.

Separation of Two Convex Sets:

Given two convex sets K and C , if K is compact, C is closed, and $K \cap C = \emptyset$, then they are strictly separable such that $a^T x < b < a^T y$ for all $x \in K$ and $y \in C$.

Analytical Extension on the Compactness Requirement:

The compactness condition is strictly binding to prevent asymptotic convergence. Consider the closed, disjoint sets $C = \{(x, y) \in \mathbb{R}^2 : y \geq e^x\}$ and $K = \{(x, y) \in \mathbb{R}^2 : y \leq 0\}$. Lacking compactness, these sets converge asymptotically as $x \rightarrow -\infty$, rendering strict separation via a hyperplane mathematically impossible.

The notation in the notes uses the symbol x in two distinct roles, which causes the ambiguity:

- In the condition $a^T x < b < a^T y, \forall x \in C$:** Here, x is a **dummy variable representing every point inside the set C** . It lives entirely inside the circular blob labeled C . The strict inequality $a^T x < b$ means that the entire set C lies strictly on one side of the hyperplane (in the open lower half-space).
- In the hyperplane definition $a^T x = b$:** Here, x is the **coordinate variable describing the hyperplane itself**. It lives along the dividing line. Every point on that line satisfies $a^T x = b$.

To avoid notation collision, the separation condition can be written as:

$$a^T c < b < a^T y \quad \forall c \in C$$

while the separating hyperplane is:

$$H = \{z \in \mathbb{R}^n : a^T z = b\}$$

In the diagram:

- The points $x \in C$** lie inside the circle C on the left.
- The points satisfying $a^T x = b$** form the straight boundary line in the middle.
- The point y** lives as an isolated point on the right (in the open upper half-space $a^T y > b$).

From <https://www.youtube.com/watch?v=3G8H8H8H8H8>

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1. Why K Mus
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this:

Definition of compactness.
Collections of open balls

Compact sets are approximated by

written in the notes with strict inequalities form **open** sets, not closed ones.

Definition of Half-Spaces
A hyperplane defined by $H = \{x \in \mathbb{R}^n \mid a^T x = b\}$ with non-zero vector $a \in \mathbb{R}^n$ and scalar $b \in \mathbb{R}$:

Half-Spaces (Strict Inequalities):
Lower Half-Space: $\{x \in \mathbb{R}^n \mid a^T x < b\}$
Upper Half-Space: $\{x \in \mathbb{R}^n \mid a^T x > b\}$
Logical property: These sets exclude the boundary hyperplane H . Because the linear functional $f(x) = a^T x$ is continuous, the preimages of the open intervals $(-\infty, b)$ and (b, ∞) are strictly open sets in the standard Euclidean topology on $\mathbb{R}^n$.

Half-Spaces (Weak Inequalities):
Lower Half-Space: $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$
Upper Half-Space: $\{x \in \mathbb{R}^n \mid a^T x \geq b\}$
Logical property: These sets explicitly include their boundary, which is the hyperplane H itself. The preimages of $(-\infty, b]$ and $[b, \infty)$ under $f(x) = a^T x$ are closed sets.

[https://www.youtube.com/watch?v=...](#)

Additional requirements of compactness and closedness are directly related to the existence of a strictly positive Euclidean distance between the sets, as a separating hyperplane establishes an ordered, one-dimensional projection.

Theorem: Compact and Closed Must Be Closed
A strictly separating hyperplane ($a^T x < b < a^T y$), the minimum distance between the two sets must be strictly positive:

$$\inf\{\|x - y\| \mid x \in K, y \in C\} > 0$$

Counterexample: Suppose $K = \{0\}$ and $C = (0, 1) \subset \mathbb{R}$. Both sets are compact, and bounded, but C is open. Their distance is $\inf\{0 - y \mid y \in C\} = 0$ because the origin is a boundary point of C , no number b can satisfy $0 < b < y$ for all $y \in (0, 1)$.
Set is compact (both closed but unbounded): Disjoint sets can asymptote toward each other, causing the infimum to be equal zero. The counterexample in the diagram illustrates

Given two convex sets K & C
 if K is compact, C closed, $K \cap C = \emptyset$

then $\exists a \in \mathbb{R}^1, b \in \mathbb{R}$

$$a^T x < b < a^T y$$

$$\forall x \in K, y \in C$$

why does C have to be closed?



so what?
 guarantees separation of $x+y$? or ordering?

why K needs to be compact?
 $C = \{(x,y) | y \geq e^x\}$
 $K = \{(x,y) | y \leq 0\}$
 cannot separate w/ hyperplane due to asymptotic convergence

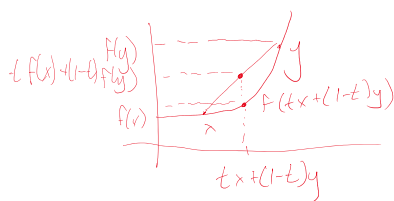
Convex & Concave Functions

convex set $C, t \in [0,1]$

$$f: C \rightarrow \mathbb{R}$$

f is convex if

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$$



convex functions always have non-convex sets under curve?

f concave if

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y)$$

& vice versa

strictly concave or convex if $>$ or $<$ when $x \neq y$ & $t \in (0,1)$

For concave functions?

Jensen's Inequality

$$x_1, x_2, \dots, x_n \in C$$

$$\lambda_1, \lambda_2, \dots, \lambda_n \geq 0$$

$$\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$$

if f concave,

convexity & concavity not preserved under monotone transformation

3. Convex & Concave Functions

A function $f: C \rightarrow \mathbb{R}$ defined on a convex set C is concave if for all $x, y \in C$ and

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y)$$

The function is convex if the inequality is reversed ($\leq$). Strictly concave or convex functions replace weak inequalities with strict inequalities for $t \in (0,1)$.

Analytical Extension on Transformations:

Concavity and convexity are cardinal properties. While they are preserved under transformations, they are generally not preserved under arbitrary monotone transformations. This necessitates the use of quasiconcavity in ordinal preference frameworks.

Jensen's Inequality:

For a concave function, the structural inequality generalizes to linear combinations

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \geq \sum_{i=1}^n \lambda_i f(x_i)$$

given $\lambda_i \geq 0$ and $\sum \lambda_i = 1$.

Convex functions do **not** always have non-convex sets under their curve, but concave functions have convex sets under their curve (their hypograph).

To formalize this, define the region under the curve (the **hypograph**) and the region above the curve (the **epigraph**):

- Hypograph (under the curve):** $\text{hyp}(f) = \{(x, y) \in C \times \mathbb{R} \mid y \leq f(x)\}$
- Epigraph (above the curve):** $\text{epi}(f) = \{(x, y) \in C \times \mathbb{R} \mid y \geq f(x)\}$

1. Concave Functions Always Have Convex Sets Under Their Curve

A function $f: C \rightarrow \mathbb{R}$ is concave if and only if its hypograph $\text{hyp}(f)$ is a convex set.

- Proof:** Let $(x_1, y_1), (x_2, y_2) \in \text{hyp}(f)$, meaning $y_1 \leq f(x_1)$ and $y_2 \leq f(x_2)$. For any $t \in [0,1]$, the convex combination:

$$t(x_1, y_1) + (1-t)(x_2, y_2) = (tx_1 + (1-t)x_2, ty_1 + (1-t)y_2)$$

By definition of concavity:

$$ty_1 + (1-t)y_2 \leq tf(x_1) + (1-t)f(x_2) \leq f(tx_1 + (1-t)x_2)$$

Thus, the combination lies in $\text{hyp}(f)$, confirming $\text{hyp}(f)$ is convex.

2. Convex Functions and the Set Under the Curve

For a convex function, its epigraph $\text{epi}(f)$ (the set above the curve) is always a convex set. However, whether the hypograph $\text{hyp}(f)$ (the set under the curve) is non-convex depends on whether the

• If neither s closed sets distance to this:

$$C = \{(u, v) \mid$$

Both sets are vertical gap separating t fails strict se

• The role of $\mathbb{R}^n$ and C is minimum or $\text{dist}(K, C) =$ strictly betw

2. Objective of The separating an algebraic

• Geometric disjoint open $\{z \mid a^T z < b$

• Dimension a^T projects line $\mathbb{R}$. The K strictly pre a :

$\sup_{x \in K} a^T x < b$
 In economic th existence of a production pos

et is compact (both closed but unbounded): Disjoint
can asymptote toward each other, causing the infimum
equal zero. The counterexample in the diagram illustrates

$$\in \mathbb{R}^2 \mid v \geq e^u\}, \quad K = \{(u, v) \in \mathbb{R}^2 \mid v \leq 0\}$$

re closed, convex, and $K \cap C = \emptyset$. However, as $u \rightarrow -\infty$, the
 $e^u - 0 \rightarrow 0$, giving $\text{dist}(K, C) = 0$. Any horizontal line
them must be $v = 0$, which touches the boundary of K and
eparation ($a^T x < b < a^T y$).

compactness: When K is compact (closed and bounded in
closed, the continuous distance function achieves a
 $K \times C$ by the Extreme Value Theorem. This guarantees
 $\delta > 0$, providing the necessary gap to place a hyperplane
een them.

of the Separating Hyperplane: Separation vs. Ordering
g hyperplane accomplishes both **geometric separation** and
linear ordering:

Separation: It partitions the n -dimensional space into two
n half-spaces, isolating K completely in the lower half-space
and C completely in the upper half-space $\{z \mid a^T z > b\}$.

ality Reduction and Ordering: The mapping $\pi(z) =$
vectors from $\mathbb{R}^n$ onto a 1-dimensional continuum on the real
scalar b acts as a cut-off point, ensuring every element of
cedes every element of C along the normal vector direction

$$< \inf_{y \in C} a^T y$$

theory, this 1-dimensional projection guarantees the
linear pricing rule or Lagrange multiplier vector a that orders
ssibilities and consumption preferences.

[problem 10.10](#)

$t \in [0,1]$:

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Equation

$$f(\lambda x) \geq \lambda f(x)$$

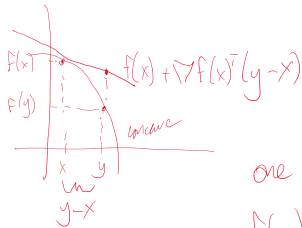
Quasiconcavity survives
monotone transformation

First order characterization of concavity

$f: C \rightarrow \mathbb{R}$ differentiable

then f is concave

$$\Leftrightarrow f(y) \leq f(x) + \nabla f(x)^T(y-x)$$



one dimension

$$f(x) + f'(x)(y-x)$$

why use f' ?

if f concave + differentiable

$$\nabla f(x^*) = 0 \Rightarrow x^*$$

critical point global max

$$\text{since } f(y) \leq f(x^*) + \nabla f(x^*)^T(y-x)$$

$$f(y) \leq f(x^*)$$

for all $y \in C$

Proof for if & only if above:

$\Rightarrow$ suppose f concave

$$x, y \in \mathbb{R}^n \quad t \in \mathbb{R}$$

$$\text{let } f(tx + (1-t)y) \geq t f(x) + (1-t)f(y)$$

$$\text{Define } g(t) = f(x + t(y-x))$$

$$g(0) = f(x)$$

$$g(1) = f(y)$$

f concave implies

$$g(t) \geq (1-t)f(x) + t f(y)$$

$$(1-t)g(0) + t g(1)$$

does it matter if tx or $(1-t)x$?
can we swap variables as below?

whether the hypograph $\text{hyp}(f)$ (the set under the curve) is non-convex depends on whether the function has strict curvature:

- **Strictly Convex Functions (Non-Convex Hypograph):** If f has strict curvature (such as $f(x) = e^x$), the hypograph $\text{hyp}(f)$ is **non-convex**. For example, for $f(x) = x^2$, the points $(-1, 1)$ and $(1, 1)$ lie on the boundary of the hypograph, but their midpoint $(0, 1)$ violates the condition since $f(0) = 0$.
- **Affine Functions (Convex Hypograph):** If f is affine (e.g., $f(x) = ax + b$), it is simultaneously convex and concave. In this case, the hypograph $\text{hyp}(f) = \{(x, y) \mid y \leq ax + b\}$ is a half-space, which is a **convex set**.

4. First-Order Characterization of Concavity

For a differentiable function $f: C \rightarrow \mathbb{R}$, f is concave if and only if for all $x, y \in C$:

$$f(y) \leq f(x) + \nabla f(x)^T(y-x)$$

Analytical Extension on Global Optimization

The right-hand side represents a global affine approximation to f at x . Evaluating this at a critical point x^* where the gradient $\nabla f(x^*) = 0$ simplifies the expression to $f(y) \leq f(x^*)$ for all $y \in C$. This verifies x^* as a global maximum, providing a mathematical underpinning for unconstrained optimization.

Proof Outline for Necessity:

Define the restriction mapping $g(t) = f(x + t(y-x))$ for $t \in [0, 1]$. Then $g(0) = f(x)$ and $g(1) = f(y)$. If f is concave, it implies $g(t)$ is concave on $[0, 1]$. Evaluating $g(t)$ at the boundary points gives $g(0) = f(x)$ and $g(1) = f(y)$. Taking the limit as $t \rightarrow 0$ yields the directional derivative $\nabla f(x)^T(y-x) \geq 0$, establishing the theorem.

Swapping the weights between tx and $(1-t)x$ does not alter the validity or direction of the inequality. This parameter symmetry over the unit interval $[0, 1]$ is a key property of the convex combination.

Symmetry of the Convex Combination

The definition of concavity holds for all $x, y \in C$ and all $t \in [0, 1]$:

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y)$$

Defining a complementary parameter $s = 1 - t$, notice that as t traverses $[0, 1]$, s also traverses $[0, 1]$ in the opposite direction. Substituting s directly yields:

$$f((1-s)x + sy) \geq (1-s)f(x) + sf(y)$$

Because the variable labels x and y and the indices t and s are dummy variables quantified over the same domain C and unit interval $[0, 1]$, both representations define the exact same line segment:

$$\{tx + (1-t)y \mid t \in [0, 1]\} \equiv \{(1-t)x + ty \mid t \in [0, 1]\}$$

Application to the Proof in the Notes

In the notes, the restriction along the line segment is defined as:

$$g(t) = f(x + t(y-x)) = f((1-t)x + ty)$$

Evaluating $g(t)$ at the boundary points gives:

- $g(0) = f(x)$
- $g(1) = f(y)$

Applying concavity directly with weights $(1-t)$ on x and t on y :

$$g(t) = f((1-t)x + ty) \geq (1-t)f(x) + tf(y) = (1-t)g(0) + tg(1)$$

Subtracting $g(0)$ from both sides:

$$g(t) - g(0) \geq -tg(0) + tg(1) = t(g(1) - g(0))$$

Dividing by $t > 0$ preserves the inequality direction:

$$\frac{g(t) - g(0)}{t} \geq g(1) - g(0)$$

Taking the limit as $t \rightarrow 0^+$ preserves the weak inequality and yields:

$$g'(0) \geq g(1) - g(0) \Rightarrow \nabla f(x)^T(y-x) \geq f(y) - f(x)$$

Rearranging terms produces the first-order overestimator:

$$f(y) \leq f(x) + \nabla f(x)^T(y-x)$$

e function

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$$g(t) \geq (1-t)f(x) + tf(y) \\ (1-t)g(0) + tg(1)$$

$$\frac{g(t) - g(0)}{t} \geq (g(1) - g(0))$$

looks like derivative

Concave implies differentiable?

Let $t \rightarrow 0$

$$\text{then } g'(0) \geq g(1) - g(0)$$

$$g'(0) = Df(t) = Df(y-x) \\ = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) (y-x) \\ = \nabla f(x)^T (y-x) \geq f(y) - f(x)$$

$$f(x) + \nabla f(x)^T (y-x) \geq f(y)$$

is this only

for x^*
crit point?
or all x ?

$$\Leftarrow \text{Let } z = (1-t)x + ty$$

$$f(x) \leq f(z) + \nabla f(z)^T (x-z)$$

$$(1) \quad (1-t)$$

$$f(y) \leq f(z) + \nabla f(z)^T (y-z)$$

$$(2) \quad (t) \quad \text{add } (1)(1-t) \\ \downarrow + (2)(t)$$

$$(1-t)f(x) + tf(y) \leq f(z) + \nabla f(z)^T [(1-t)(x-z) + t(y-z)]$$

Concavity does not imply differentiability.

A standard counterexample on $\mathbb{R}$ is the function $f(x) = -|x|$, any interval containing 0 (or concave on $\mathbb{R}$), yet fails to be differentiable at 0.

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = -1 \quad \text{and} \quad \lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = 1$$

Analytical Regularity of Concave Functions

While concavity does not ensure differentiability everywhere, properties on the function:

- **Continuity on Open Domains:** If $f: U \rightarrow \mathbb{R}$ is concave (or convex) on $U \subseteq \mathbb{R}^n$, then f is locally Lipschitz continuous on U .
- **One-Sided Derivatives ($n = 1$):** For a concave function f : derivative $f'_-(x)$ and right derivative $f'_+(x)$ exist everywhere.

$$f'_-(x) \geq f'_+(x) \geq f'_-(y) \geq f'_+(y) \quad \text{for all } a < x < y < b$$

Both f'_- and f'_+ are monotonically non-increasing.

- **Almost Everywhere Differentiability:** Because a monotone function has at most countably many jump discontinuities, the set of non-differentiability is at most countable and has Lebesgue measure zero.

$$D = \{x \in (a, b): f'_+(x) \neq f'_-(x)\}$$

By classical convex analysis results, every concave function on $U \subseteq \mathbb{R}^n$ is differentiable almost everywhere with respect to the Lebesgue measure.

- **Superdifferentials:** In nonsmooth analysis and convex optimization, the superdifferential $\partial f(x)$ of a concave function $f: U \rightarrow \mathbb{R}$ is defined as:

$$\partial f(x) = \{v \in \mathbb{R}^n: f(y) \leq f(x) + v^T(y-x), \forall y \in U\}$$

$\partial f(x)$ is a nonempty, compact, and convex set for every x in the interior of U . f is differentiable at x if and only if $\partial f(x)$ is a singleton, in which case $\partial f(x) = \{\nabla f(x)\}$.

The proof holds for all $x \in C$, not merely at a critical point x^* .

Analytical Framework

The first-order condition defines a global property across the entire domain C .

$$f(y) \leq f(x) + \nabla f(x)^T (y-x) \quad \forall x, y \in C$$

- **Arbitrary $x \in C$:** The affine function $T(y) = f(x) + \nabla f(x)^T (y-x)$ describes the supporting hyperplane at evaluation point x . The inequality establishes that for any concave function, any point sits globally above or on the function surface.
- **Special Case at Critical Point x^* :** If $x = x^*$ happens to be a critical point, the term vanishes:

$$f(y) \leq f(x^*) + 0^T (y-x^*) \Rightarrow f(y) \leq f(x^*) \quad \forall y \in C$$

This confirms that any stationary point of a differentiable concave function is a global maximizer.

The construction of $g(t) = f(x + t(y-x))$ relies solely on the arbitrary choice of x and y .

which is strictly concave on
differentiable at $x = 0$:

it imposes strong regularity

convex) on an open convex set

$(a, b) \rightarrow \mathbb{R}$, both the left
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ice of $x, y \in C$, without requiring

$$(1-t)f(x) + t f(y) \leq f(z) + \underbrace{\nabla f(z)^T \left[(1-t)(x-z) + t(y-z) \right]}_{\text{equals 0}}$$

s.t.

$$(1-t)f(x) + t f(y) \leq f(z) = f((1-t)x + ty)$$

Second order characterization of concavity

for $f \in C^2(C)$ twice cont
differentiable
Hessian symmetric

f is concave

$\Rightarrow$ Hessian negative
(semi) definite

$$H_f \preceq 0$$

note:

$$H_f \succeq 0$$

$\Rightarrow f$ strictly
concave



converse

not generally
true

strict concave
does not imply
negative def. H

for some $g(t) = f(x + t(y-x))$

take $g''(t)$ +

evaluate Hessian

5. Second-Order Characterization of Concavity

For a twice continuously differentiable function $f \in C^2(C)$, evaluate the symmetric Hessian matrix. The function is concave if and only if the Hessian is negative semi-definite ($H \leq 0$).

Analytical Extension on Strict Concavity:

A strictly negative definite Hessian implies strict concavity. However, the converse does not hold universally. For instance, the function $f(x) = -x^4$ is strictly concave globally, yet its Hessian evaluated at the origin evaluates to zero, failing the strict negative definiteness test. Assessing definiteness formally requires evaluating the determinant of principal minors or the matrix eigenvalues.

How to rigorously establish the second-order conditions for concavity, analyze the symmetric Hessian matrix $H(x)$ of a twice continuously differentiable function $f \in C^2(\mathbb{R}^n)$. Negative definiteness of $H(x)$ guarantees strict concavity, while negative semi-definiteness guarantees weak concavity. Consider the multivariate function $f(x, y) = -x^2 - y^2 + xy$. The corresponding constant Hessian matrix is:

$$\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

How is the mechanical application of the three primary analytical tests to determine the definiteness of this matrix.

The Eigenvalue Test

Spectral theory dictates that a real symmetric matrix is diagonalizable and possesses strictly real eigenvalues.

Condition: H is negative definite if all eigenvalues satisfy $\lambda_i < 0$. H is negative semi-definite if all $\lambda_i \leq 0$.

Application: We solve the characteristic equation $\det(H - \lambda I) = 0$.

$$(-2 - \lambda)^2 - 1 = \lambda^2 + 4\lambda + 3 = 0$$

Factoring the characteristic polynomial yields $(\lambda + 3)(\lambda + 1) = 0$. The roots are $\lambda_1 = -1$ and $\lambda_2 = -3$.

Result: Because all eigenvalues are strictly negative, H is strictly negative definite, confirming $f(x, y)$ is strictly concave.

The Quadratic Form Test

The quadratic form evaluates the structural curvature of the function along an arbitrary non-zero directional vector $v \in \mathbb{R}^n$.

Condition: H is negative definite if $v^T H v < 0$ for all $v \neq 0$.

Application: Let $v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. The quadratic form evaluates to:

$$Hv = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = -2v_1^2 + 2v_1v_2 - 2v_2^2$$

Completing the square allows the algebraic isolation of terms:

evaluate Hessian
eigenvalue test
& quadratic test?
det of principal minors
test?

negative det. H
ex: $f(x) = -x^4$

Quasi concavity

$$f: C \rightarrow \mathbb{R}$$

$$\text{if } f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$$

start if $x \neq y, t \in (0, 1)$

concave
 $\Rightarrow$ quasiconcave?

upper contour set

$$U_d = \{x : f(x) \geq d\}$$

f is quasiconcave

$\Rightarrow U_d$ is convex

$$Hv = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = -2v_1^2 + 2v_1v_2 - 2v_2^2$$

Completing the square allows the algebraic isolation of terms:

$$v_1^2 - v_2^2 - (v_1 - v_2)^2$$

Result: Since v is non-zero, at least one of the squared terms is strictly positive. Multiplying by the negative constant ensures the entire expression is strictly less than zero. H is negative definite.

The Principal Minor Determinants Test

This test leverages Sylvester's Criterion by evaluating the sub-determinants of the Hessian matrix.

Condition for Negative Definiteness: The leading principal minors D_k must alternate in sign, strictly beginning with a negative sign: $(-1)^k D_k > 0$.

Condition for Negative Semi-Definiteness: All principal minors (not just leading minors) of order k , denoted Δ_k , must satisfy $(-1)^k \Delta_k \geq 0$.

Application: For the 2×2 matrix H :

$$D_1 = -2 \text{ (negative)}$$

$$D_2 = \det(H) = (-2)(-2) - (1)(1) = 3 \text{ (positive)}$$

Result: The sequence of signs is negative, then positive. The strict alternation condition is satisfied, confirming negative definiteness.

<<https://gemini.google.com/app/264b755bb013b5de>>

6. Quasiconcavity

A function $f: C \rightarrow \mathbb{R}$ is quasiconcave if for all $x, y \in C$ and $t \in [0, 1]$:

$$f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$$

Alternatively, defining the upper contour

set as $U_\alpha = \{x: f(x) \geq \alpha\}$, f is

quasiconcave if and only if U_α is a convex

set for all α . While concavity implies

quasiconcavity, the converse is false.

Analytical Extension on Monotone Transformations:

Unlike concavity, quasiconcavity is an ordinal property strictly preserved under monotonically increasing

transformations. If f is quasiconcave and

ϕ is monotonically increasing, the

composite mapping $\phi(f(x))$ retains

quasiconcavity. This topological property

allows continuous mapping of convex

preference sets to functional utility

representations.

Strict Quasiconcavity and Optimization:

Strict quasiconcavity enforces strict

inequality $f(tx + (1-t)y) >$

$\min\{f(x), f(y)\}$. If a maximum exists on

a convex domain, strict quasiconcavity

guarantees it is unique, ruling out a flat

plateau of global maxima. The existence

of a maximum in bounded optimizations

is independently guaranteed by the

Weierstrass extreme value theorem

when operating on a compact domain.

$\Leftarrow \Rightarrow U_\alpha$ is convex
 $\forall \alpha$

$\Rightarrow$ Take $x, y \in U_\alpha$

$$f(x) \geq \alpha$$

$$f(y) \geq \alpha$$

want to show

$$f(tx + (1-t)y) \geq \alpha$$

s.t. $tx + (1-t)y \in U_\alpha$

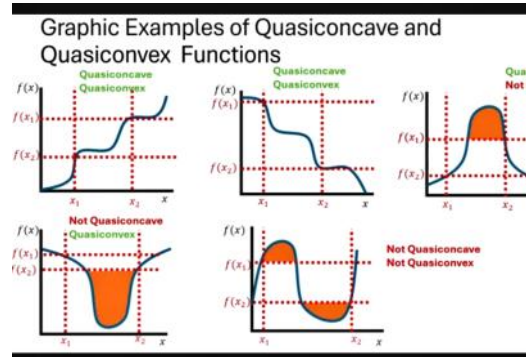
by quasi concavity

$$f(tx + (1-t)y) \geq \min(f(x), f(y)) \geq \alpha$$

$\Leftarrow$ If $f(x) \geq \alpha, f(y) \geq \alpha$

want to show that $f(tx + (1-t)y) \geq \alpha$

$$\text{Choose } \alpha = \min \{f(x), f(y)\}$$



iconcave
Quasiconvex

$$\text{then } f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$$

since $t \geq 0$,
inequality holds by def

Quasiconcave lets us take
monotone transformation
(Quasiconcavity preserved
under monotone transforms)

Suppose f quasiconcave

$$f: C \rightarrow \mathbb{R}$$

$$\text{take } \phi: f(C) \rightarrow \mathbb{R}$$

w/ ϕ monotonically increasing

Then $\phi(f(x))$ is quasiconcave

proof:

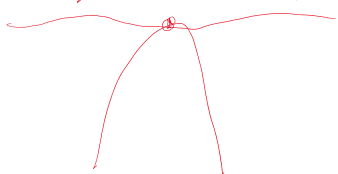
Let $x, y \in C$

proof:

$$\begin{aligned}\phi(f(tx + (1-t)y)) &\geq \\ &\phi(\min\{f(x), f(y)\}) \\ &= \min\{\phi(f(x)), \phi(f(y))\}\end{aligned}$$

Strict Quasiconcavity
to identify unique max

strict q-concave
 $\Rightarrow$ max unique



proof:

take $x \neq y$, both maximizers

$$f(x) = f(y) = M = \max_{z \in C} f(z)$$

Strict Quasiconcavity

$$f(tx + (1-t)y) > \min\{f(x), f(y)\}$$

$$f(x) + f(1-x)y) = \max_{z \in C} f(z)$$

$$= M$$

$$= \max_{z \in C} f(z)$$

then contradiction, so $x=y$
□

under strict q-concavity,
maximizer must be unique

Existence of max guaranteed
by compact sets

but unique max requires
strict quasiconcavity

usually in practice we
check strict concavity

If $f \prec 0 \Rightarrow$ strict
concavity $\Rightarrow$ strict
quasiconcavity

does weak
concavity
imply
weak
quasiconcavity?
or only
strict concavity
implies strict
quasiconcavity?

Real Analysis: Convexity, Hyperplanes, and Optimization
1. Convex Sets
A set $S \subset \mathbb{R}^n$ is defined as convex if for any $x, y \in S$
and $\lambda \in [0, 1]$, the combination $\lambda x + (1-\lambda)y \in S$.
Example: The closed sphere defined by $S = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ is a convex set.
Analytical Extension on Linear vs. Convex Combinations:
A convex combination of a finite set of points is expressed as $\sum \lambda_i x_i$ subject to the simplex constraints $\sum \lambda_i = 1$ and $\lambda_i \geq 0$. This restricts the resultant vector to the geometric simplex defined by the basis vectors, whereas a general linear combination lacks these bounds and can span the entire vector space.
Convex Hull:
For any arbitrary set $S \subset \mathbb{R}^n$, the convex hull $\text{co}(S)$ is defined as the set of all convex combinations of points within S . If S is convex, then $\text{co}(S) = S$.
Analytical Extension on Topological Properties:
The convex hull represents the minimal convex set containing S . Open sets can satisfy convexity (e.g., an open ball). Furthermore, compactness (being closed and bounded in $\mathbb{R}^n$) is structurally independent of convexity; a set can be convex without being compact, and vice versa.
2. Separating Hyperplanes
A hyperplane takes the linear algebraic form $\{x \in \mathbb{R}^n : a^T x = b\}$, with $a \in \mathbb{R}^n$

Yes, weak concavity strictly implies weak quasiconcavity. The relationship holds for both the weak and strict forms of the definitions. The analytical proof for weak concavity implying weak quasiconcavity relies on straightforward algebraic bounding.

Proof of Implication (Weak)

Assume a function $f: C \rightarrow \mathbb{R}$ is weakly concave on a convex set C . By definition, for any $x, y \in C$ and any scalar $t \in [0,1]$:

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y)$$

To evaluate quasiconcavity, we must establish a relationship with $\min\{f(x), f(y)\}$. Without loss of generality, assume that $f(x) \geq f(y)$. Consequently, $\min\{f(x), f(y)\} = f(y)$.

Because $f(x) \geq f(y)$ and $t \geq 0$, we can bound the right side of the concavity definition from below by substituting $f(y)$ for $f(x)$:

$$tf(x) + (1-t)f(y) \geq tf(y) + (1-t)f(y)$$

Simplifying the right side of this new inequality yields:

$$tf(y) + (1-t)f(y) = f(y)$$

By transitivity, we connect the original concavity condition to this lower bound:

$$f(tx + (1-t)y) \geq f(y)$$

Since $f(y) = \min\{f(x), f(y)\}$, this results in the exact formal definition of weak quasiconcavity:

$$f(tx + (1-t)y) \geq \min\{f(x), f(y)\}$$

Extension to Strict Concavity

The same mechanical structure applies to strict concavity implying strict quasiconcavity.

If f is strictly concave, then for any $x \neq y$ and $t \in (0,1)$:

$$f(tx + (1-t)y) > tf(x) + (1-t)f(y)$$

Maintaining the assumption that $f(x) \geq f(y)$, the same substitution provides:

$$tf(x) + (1-t)f(y) \geq f(y)$$

Combining the strict inequality of the premise with the weak inequality of the substitution yields a strict inequality in the conclusion:

$$f(tx + (1-t)y) > \min\{f(x), f(y)\}$$

Therefore, the structural hierarchy maps consistently: weak concavity guarantees weak quasiconcavity, and strict concavity guarantees strict quasiconcavity.

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